

	Case Name: Nuclear Healthcare	Sector	Construction (civil)
	OR-AS Operations Research - Applications and Solutions www.or-as.be info@or-as.be	Baseline Schedule	Schedule with resources
			Schedule with costs
		Risk Analysis	Random simulation
			One of nine std. scenarios
Submitted by	Jens Martens		User defined distributions
Date	2019	Project Control	Automatic tracking
File Name	C2019-04 Nuclear Healthcare		Tracking based on user input

1. Project description

Project authenticity

2. Project properties

2.1. Baseline Schedule

General		Network topology	
# Activities	38	Serial/Parallel (SP)	84%
Planned Duration (PD)	373	Activity Distribution (AD)	85%
Budget At Completion (BAC)	4,318,950.00€	Length of Arcs (LA)	60%
Renewable Resources	-	Topological Float (TF)	2%
Consumable Resources	-		

* standard eight-hour working days

2.2. Risk Analysis

Random simulation by ProTrack was performed using the default symmetric triangular risk distribution profiles.

	Cost sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	8	17	3.50
CRI-rho	28	24	0.40
CRI-tau	46	46	0.31

	Time sensitivity		
	avg [%]	std dev [%]	skew [-]
CI	3	11	4.88
SI	18	23	1.92
SSI	4	15	5.31
CRI-r	10	16	4.56
CRI-rho	10	16	4.68
CRI-tau	10	14	4.21

	Resource sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	N/A	N/A	N/A
CRI-rho	N/A	N/A	N/A
CRI-tau	N/A	N/A	N/A

2.3. Project Control

2.3.1. Simulated forecasting accuracy

The accuracy of time and cost forecasting methods has been evaluated based on Monte Carlo simulation runs using the risk profiles described in section “2.2. Risk Analysis”. Based on these risk profiles, the Mean Absolute Percentage Error (MAPE) and Mean Percentage Error (MPE) has been calculated to evaluate the expected accuracy of the time and cost predictions, EAC(t) and EAC, respectively.

Simulated EAC(t) accuracy			Simulated EAC accuracy		
method - PF	MAPE [%]	MPE [%]	method (PF)	MAPE [%]	MPE [%]
PV - 1	N/A	N/A	1	N/A	N/A
PV - SPI	N/A	N/A	CPI	N/A	N/A
PV - SCI	N/A	N/A	SPI	N/A	N/A
ED - 1	N/A	N/A	SPI(t)	N/A	N/A
ED - SPI	N/A	N/A	SCI	N/A	N/A
ED - SCI	N/A	N/A	SCI(t)	N/A	N/A
ES - 1	N/A	N/A	0.8 CPI + 0.2 SPI	N/A	N/A
ES - SPI(t)	N/A	N/A	0.8 CPI + 0.2 SPI(t)	N/A	N/A
ES - SCI(t)	N/A	N/A			

According to the MAPE values¹ the best performance for time forecasting can be expected from the unweighted Earned Schedule method. For cost forecasting the unweighted and CPI-weighted methods should yield the best results.

2.3.2. Tracking description

Tracking authenticity

Manual tracking was performed over 24 tracking periods. The Real Duration and Real Cost mentioned in section “2.3.3. Earned Value Management” are based on manual user input.

The tracking information obtained from the project owner and introduced in ProTrack includes actual activity start dates, durations and costs.

¹ The MAPE gives the best indication for the forecast accuracy (the lower the MAPE, the more accurate the method) since all deviations from the targeted real duration (real cost) are cumulated, whereas for the MPE underestimates can be compensated by overestimates and vice versa, possibly leading to an overly positive evaluation of a certain method. However, the MPE can provide useful information about the nature of the deviations, i.e. does the method rather underestimate or overestimate the real duration (real cost)?

2.3.3. Earned Value Management

2.3.3.1. Performance metrics

	CV [€]	SV [€]	SV(t) [d]	CPI [-]	SPI [-]	SPI(t) [-]	p-factor [-]
avg	61675.44	-893333.46	-64.71	1.13	0.91	0.88	0.93
std dev	22845.26	1033226.54	73.72	0.14	0.86	0.24	0.12
final	86396.00	0.00	-146	1.02	1.00	0.72	1.00

2.3.3.2. Time forecasting

PD	373 d	Real Duration	520 d	Late/early	39.41%
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EAC(t)			Real Accuracy	
method - PF	avg [d]	std dev [d]	MAPE [%]	MPE [%]
PV - 1	450.58	89.28	19.78	13.35
PV - SPI	563.08	250.44	39.23	-8.29
PV - SCI	522.08	262.42	41.38	-0.40
ED - 1	486.08	119.50	20.63	6.52
ED - SPI	600.21	273.34	43.02	-15.42
ED - SCI	572.29	284.19	44.42	-10.06
ES - 1	438.08	73.68	17.61	15.75
ES - SPI(t)	455.17	113.37	20.42	12.47
ES - SCI(t)	431.96	134.57	24.50	16.93

2.3.3.3. Cost forecasting

BAC	4,318,950.00€	Real Cost	4,232,553.41€	Over/under budget	2%
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EAC			Real Accuracy	
method (PF)	avg [€]	std dev [€]	MAPE [%]	MPE [%]
1	4257274.67	22364.29	0.65	-0.58
CPI	3880857.89	433335.17	8.31	8.31
SPI	5559732.63	1952454.72	41.95	-31.36
SPI(t)	4685258.23	839503.33	18.24	-10.70
SCI	5170558.62	2071494.44	43.67	-22.16
SCI(t)	4326324.05	1102765.97	22.22	-2.22
0.8 CPI + 0.2 SPI	4031341.41	603415.96	10.42	4.75
0.8 CPI + 0.2 SPI(t)	3998131.49	462801.47	8.78	5.54